

07 Changes in Concentration Due to Conversion

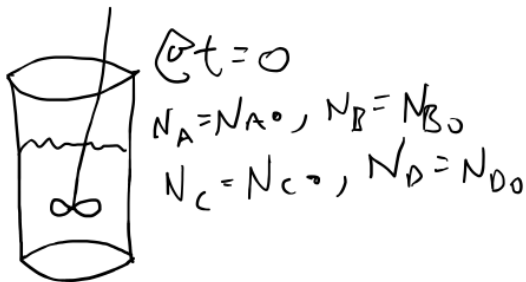
In this lecture, we'll write concentration in terms of conversion.
Concentration is written as:

- In a flow reactor, $C_i = \frac{F_i}{v}$
- In a batch reactor, $C_i = \frac{N_i}{V}$

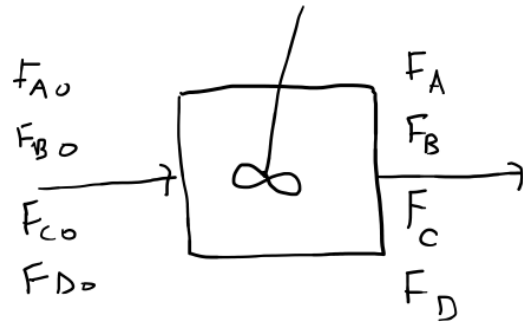
Conversion can influence F , N , V , and v .

First, let's recall the influence that X has on F and N . This is review from Material and Energy Balances:

Consider a reactor carrying out the reaction $aA + bB \rightarrow cC + dD$:



$@t=t$ (reaction has been carried out for some time)
 $N_A = N_A, N_B = N_B, N_C = N_C$
 $N_D = N_D$



Let's say that species A is the limiting reactant.

- Then, we can solve the mole balances for these reactors in terms of X_A .

$$X_A = \frac{N_{A0} - N_A}{N_{A0}} \quad \text{or} \quad \frac{F_{A0} - F_A}{F_{A0}}$$

- We can also write values of N and F for species B, C, and D in terms of X_A :

$$N_i = N_{i0} - \left(\frac{v_i}{v_A} \right) N_{A0} X_A \quad F_i = F_{i0} - \left(\frac{v_i}{v_A} \right) F_{A0} X_A$$

Recall: stoichiometric coefficients are negative for reactants and positive for products

For more on this, see Tables 4-1 and 4-3 in Fogler and also watch the video in today's module on constructing stoichiometric tables. See also Fogler Example Problem 4-1.

Re-arranging these formulas:

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$$N_i = \left[\frac{N_{i0}}{N_{A0}} - \left(\frac{\nu_i}{\nu_A} \right) X_A \right] N_{A0} \quad F_i = \left[\frac{F_{i0}}{F_{A0}} - \left(\frac{\nu_i}{\nu_A} \right) X_A \right] F_{A0}$$

$$\Theta_i = \frac{N_{i0}}{N_{A0}} \quad \text{or} \quad \frac{F_{i0}}{F_{A0}} \quad \text{where } A \text{ is the limiting reactant (LR)}$$

Important quantity/definition: $\frac{N_{i0}}{N_{A0}}$ or $\frac{F_{i0}}{F_{A0}} = \Theta_i$

$$\Theta_i = \frac{\text{moles of species } i \text{ initially or moles of species } i \text{ fed}}{\text{moles of species A initially or moles of species A fed}}$$

Hence, $N_i = N_{A0} \theta_i \left(\frac{v_i}{v_A} \right) N_{A0} X_A$ and $F_i = F_{A0} \theta_i \left(\frac{v_i}{v_A} \right) F_{A0} X_A$, where v are stoichiometric coefficients.

Important note: For proper accounting, stoichiometric coefficients are negative for reactants and positive for products

negative for reactants and positive for products.

Another important note:

- We can always write conversion for any of the reactant species, even if they are not the limiting reactant. For example, $X_A = \frac{F_{A0} - F_A}{F_{A0}}$, and $X_B = \frac{F_{B0} - F_B}{F_{B0}}$. These quantities tell us the fraction of species A and B, respectively, that has reacted.
- However, if we want to solve a mole balance in terms of X , we must use the conversion of the limiting reactant.
- We write most of our definitions and balances in this class in terms of species A ... However, species A isn't always the limiting reactant. Always use the limiting reactant as your basis of calculation when solving mole balances in terms of X .

Now let's see how X influences V and v .

Consider the gas phase reaction $A \rightarrow B + C$, and for the sake of illustration, imagine that this reaction is being carried out in a balloon:



The balloon has non-rigid walls. Assume that the reaction is carried out isothermally and that the reaction goes to completion.

The balloon is essentially a constant pressure batch reactor.

Let's consider the case where only species A is present initially and the reaction goes to completion, so ultimately, the balloon only contains species B and C.

At time = 0: $V_1 = \frac{N_1 R T_1}{P_1}$ $N_1 = \text{moles of A initially } (N_{A0})$

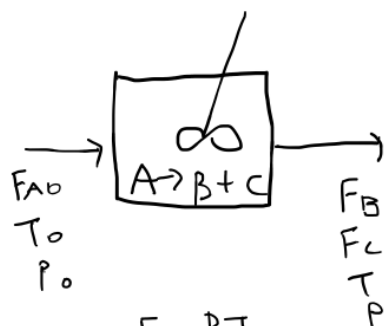
When the reaction has gone to completion:

$V_2 = \frac{N_2 R T_2}{P_2}$ $N_2 = \text{total number of moles of B \& C that were generated } (N_B + N_C)$

$N_2 = 2N_1$, $T_2 = T_1 = T$, $P_2 = P_1 = P$.

$V_2 = 2V_1$

Let's look at the analogous circumstance in a flow reactor:



Gas phase reaction,
isothermal,
isobaric, reaction goes to
completion.

$$v_0 = \frac{F_{A0} R T_0}{P_0}$$

$$v_0 = \frac{F_{A0} R T}{P}$$

$$v = \frac{(F_B + F_C) R T}{P}$$

$$T = T_0$$

$$P = P_0$$

$$F_B + F_C = 2F_{A0}$$

$$v = 2v_0$$

Key point: For gas phase reactions, the volume or volumetric flowrate changes when the total number of moles in a reaction changes.

- To dig deeper into this topic, see the related video by LearnChemE in the Canvas Module for this topic.

Key point: For liquid phase reactions, volume and volumetric flowrate can be assumed constant, as long as the system is not a semibatch system, there are no phase changes, volume expansion/contraction due to mixing can be assumed negligible, etc.

- In this class, we can always assume constant volume/volumetric flowrate in liquid reaction conditions except when the reactor is a semibatch reactor.

Summary of concentration equations:

liquid phase: $C_j = C_{LR,0} \left(\theta_j - \frac{r_j}{r_{LR}} X \right)$

gas phase: $C_j = \frac{C_{LR,0} \left(\theta_j - \frac{r_j}{r_{LR}} X \right)}{1 + \epsilon X} \left(\frac{T_0}{T} \right)$

gas phase
in terms of pressure: $P_j = \frac{P_{LR,0} \left(\theta_j - \frac{r_j}{r_{LR}} X \right)}{1 + \epsilon X}$

$$\theta_j = \frac{F_{j,0}}{F_{LR,0}}$$

r_j are stoichiometric coefficients,
negative for reactants; positive for
products

$$\epsilon = Y_{LR,0} \delta, \quad \delta = \frac{\sum_{\text{products}} |r_j| - \sum_{\text{reactants}} |r_j|}{|r_{LR}|}$$

y is mole fraction, so $Y_{LR,0} = \frac{F_{LR,0}}{F_{T,0}}$